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one can build a single integrated circuit that contains the complete functionality of the embedded system.

While the embedded system sits on top of everything, VLSI is the core part of an embedded system as the whole system depends on VLSI design.

Roles available

Embedded engineers write the software that controls the VLSI chips. Embedded software can be any code which interacts with the hardware layer, ranging from the hardware abstraction layer, device drivers, kernel programming to application programming.

To take up embedded software development as career, you need to have a working knowledge of C, C++ and Java, with some exposure to applica-

tion software development. If you have already done some programming for an embedded system, you can expand your horizons by broadening your skill sets. Embedded software is being developed using a variety of embedded and real-time operating systems, such as embedded NT, Windows CE and RTLinux. In addition, a variety of development tools and IDEs for C, C++ and Java are available. You need to learn application development in these various environments.

To be a successful programmer, you need a good understanding of design alternatives—how to choose a processor, an operating system, a programming language so that you can develop cost-effective, reliable embedded software with minimal development time. As many of the systems become network-enabled, a working knowledge of network programming is also important. Also, you need to be aware of the diversity of application areas and the specific requirements of each.

As a hardware/board designer, you could work on reference board design, new board design, and board design derivatives and modifications on existing boards. Reference boards are used to validate the capabilities of designed silicon chips.

The embedded hardware designer is responsible for hardware design schematic, PCB layout, BOM creation, hardware board debugging and testing, and system integration testing. Desired skills are hardware design using 8-/16-/32-bit microcontrollers, microprocessor-based systems, design of medium-frequency boards, hardware design of analogue, mixed-signal and digital, and understanding/implementation of the EMI and EMC concepts to hardware design.

3.5 million jobs by 2015, but talent missing

The Indian semiconductor design industry, comprising very large-scale integrated (VLSI) design, board design and embedded software companies, has set its footprints across the country.

According to an ISA-Ernst & Young

report of 2011, this industry was about \$6.5 billion in 2009 and is expected to log a compound annual growth rate of 17.3 per cent to reach \$10.6 billion in 2012.

The report also suggests that the industry promises 3.5 million jobs by 2015. But unfortunately, the industry is grappling with the problem of talent shortage.

Embedded software segment now offers more high-value activities relating to middleware, driver design and associated applications. The main reason for increased adoption of embedded software is its use by electronics companies for increased device functionality, reduced time-to-market and reduction in costs.

Raw talent: key challenge for companies

One of the basic reasons behind the shortage of talent in the country is the significant industry-academia divide.

Several engineering universities in the country are far from understanding the requirements of the industry and train the students accordingly. This has resulted into two extremes, where on one hand the industry needs talent and on the other hand there is an abundance of engineers who have not been trained in necessary skills. Industry veterans term these engineers as 'raw talent' and training to make them productive is a key challenge for them.

Companies face a tough time trying to recruit the right set of people. There are certain portions in the entire embedded ecosystem where generic electronics knowledge will help. Hardware design is one example. But then there are certain other segments, which require a good experience in device

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